Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ Bell: _____ **Homework 1:** Multi-Step Equations

**** This is a 2-page document! ****

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $-4x - 7 - 3x + 4 = 25$

$-7x - 3 = 25$

$+3 \quad +3$

$\frac{-7x}{-7} = \frac{28}{-7}$

$\boxed{x = -4}$

2. $-4(7a + 5) = -160$

$-28a - 20 = -160$

$+20 \quad +20$

$\frac{-28a}{-28} = \frac{-140}{-28}$

$\boxed{a = 5}$

3. $3k + 2(5k - 3) = 7$

$3k + 10k - 6 = 7$

$13k - 6 = 7$

$+6 \quad +6$

$\frac{13k}{13} = \frac{13}{13}$

$\boxed{k = 1}$

4. $-79 = 7w + 3(4w - 1)$

$-79 = 7w + 12w - 3$

$-79 = 19w - 3$

$+3 \quad +3$

$\frac{-76}{19} = \frac{19w}{19}$

$\boxed{-4 = w}$

5. $9 - 4(2p - 1) = 41$

$9 - 8p + 4 = 41$

$-8p + 13 = 41$

$-13 \quad -13$

$\frac{-8p}{-8} = \frac{28}{-8}$

$\boxed{p = -\frac{7}{2}}$

6. $y - 3(2y - 7) = 76$

$y - 6y + 21 = 76$

$-5y + 21 = 76$

$-21 \quad -21$

$\frac{-5y}{-5} = \frac{55}{-5}$

$\boxed{y = -11}$

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7. $15 - 5(4c - 7) = 50$

$15 - 20c + 35 = 50$

$-20c + 50 = 50$

$-50 \quad -50$

$\frac{-20c}{-20} = \frac{0}{-20}$

$\boxed{c = 0}$

8. $-(6h + 7) + 8 = -19$

$-6h - 7 + 8 = -19$

$-6h + 1 = -19$

$-1 \quad -1$

$\frac{-6h}{-6} = \frac{-20}{-6}$

$\boxed{h = \frac{10}{3}}$

9. $-\frac{1}{3}(9x + 42) - 5x = -70$

$-3x - 14 - 5x = -70$

$-8x - 14 = -70$

$+14 \quad +14$

$\frac{-8x}{-8} = \frac{-56}{-8}$

$\boxed{x = 7}$

10. $8 = -18 + \frac{3}{8}(16 - 40n)$

$8 = -18 + 6 - 15n$

$8 = -12 - 15n$

$+12 \quad +12$

$\frac{20}{-15} = \frac{-15n}{-15}$

$\boxed{-\frac{4}{3} = n}$

11. $2(5v + 3) - 4(7v + 1) = -88$

$10v + 6 - 28v - 4 = -88$

$-18v + 2 = -88$

$-2 \quad -2$

$\frac{-18v}{-18} = \frac{-90}{-18}$

$\boxed{v = 5}$

12. $7(3r - 1) - (r + 5) = -52$

$21r - 7 - r - 5 = -52$

$20r - 12 = -52$

$+12 \quad +12$

$\frac{20r}{20} = \frac{-40}{20}$

$\boxed{r = -2}$

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